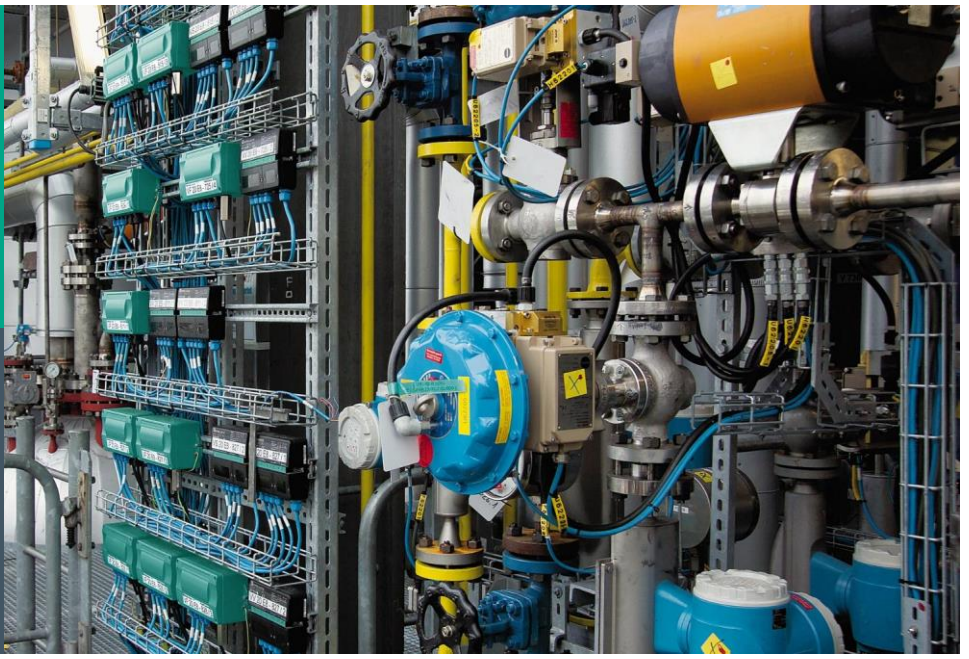
Explosion Protection

Intrinsic Safety “i”



Your automation, our passion.

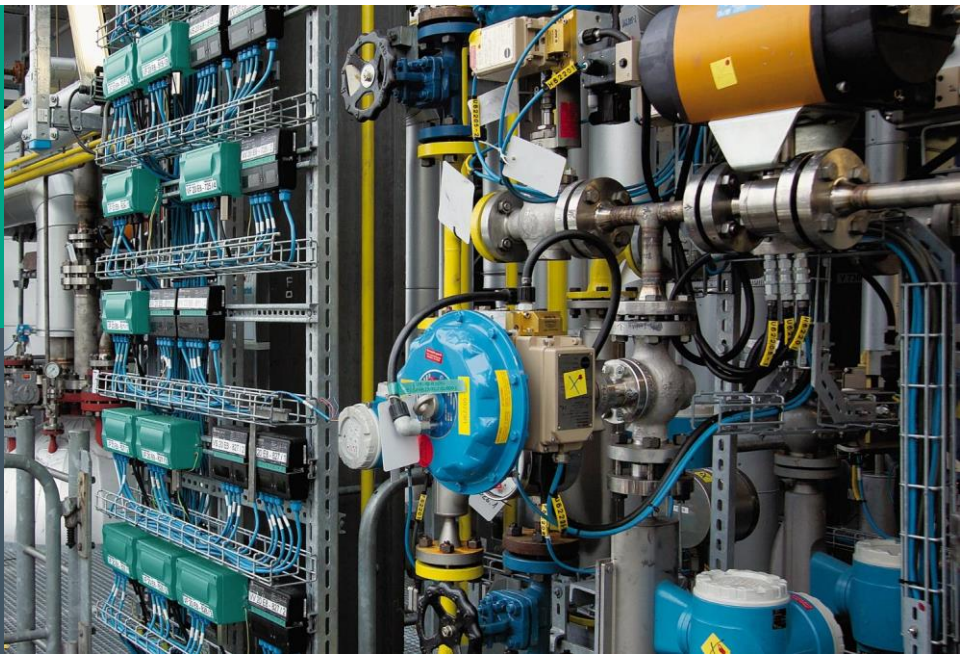
PEPPERL+FUCHS

Overview

1. **Introduction**
   Functional Principle, Terms and Definitions, Marking
2. **Selection of Equipment**
   EPL, Zones and Categories
3. **Verification of Intrinsic Safety**
   Procedure, Effects on Maintenance Work
4. **Installation Requirements**
   Cables, Terminals, Conductors, Shielding, Earthing

1. Introduction

Functional Principle, Terms and Definitions, Marking



Elements of Intrinsically Safe Circuits



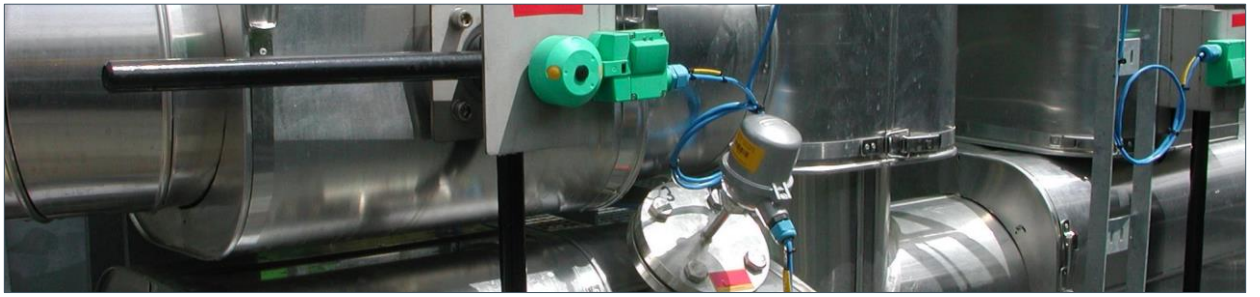
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Simple intrinsically safe circuits typically consist of:

- An associated apparatus (power supply - source)
- An intrinsically safe apparatus (field device - load)
- A cable

→ *Circuits with more than one associated apparatus will not be covered in this presentation*

Present Standards



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Present standards regarding intrinsic safety:

- EN 60079-14:2024 (equipment selection, installation, verification)
- EN 60079-11:2012 (equipment characteristics)
- EN 60079-25:2022 (intrinsically safe systems, verification)

Associated Apparatus



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Characteristics of an associated apparatus:

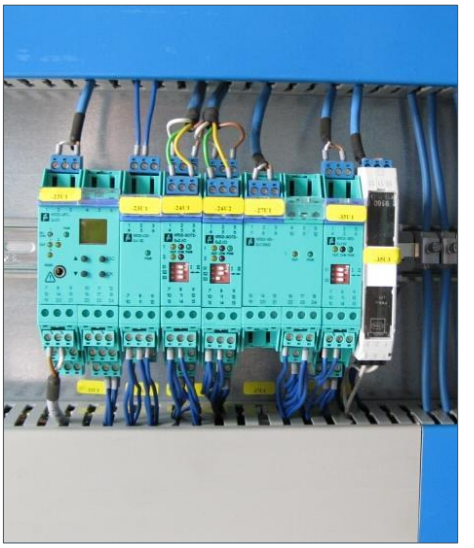
- Comprises both intrinsically safe and as well as non-intrinsically safe circuits
  - → *Typically the power supply in the cabinet in the non-hazardous area*

Associated Apparatus

II (1) G

[Ex ia Ga] IIC

- Category marked in round brackets ( )
- Type of protection marked in square brackets [ ]
- No temperature class marking
  - → *Installation in the non-hazardous area*

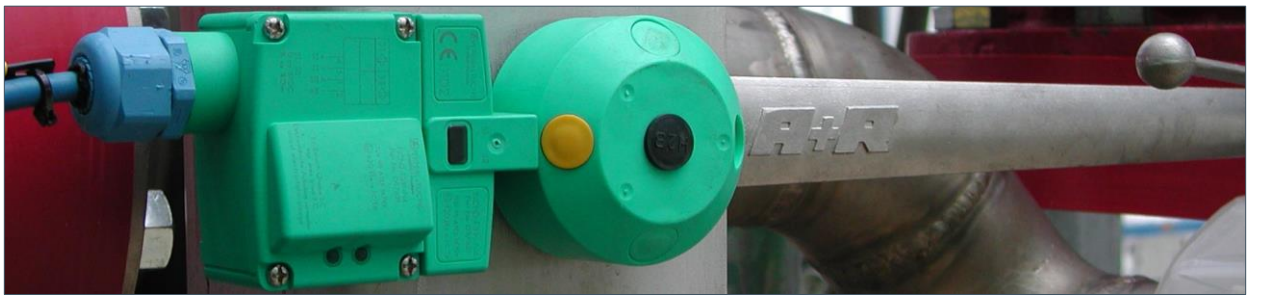


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Typically stated in the documentation:

- Max. output voltage U_o
- Max. output current I_o
- Max. output power P_o
- Max. admissible inductance L_o
- Max. admissible capacitance C_o

Intrinsically Safe Apparatus



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Characteristics of an intrinsically safe apparatus:

- Comprises intrinsically safe circuits only
  - → *E.g. sensors in the hazardous are (thermo couple, proximity sensor etc.)*

Intrinsically Safe Apparatus

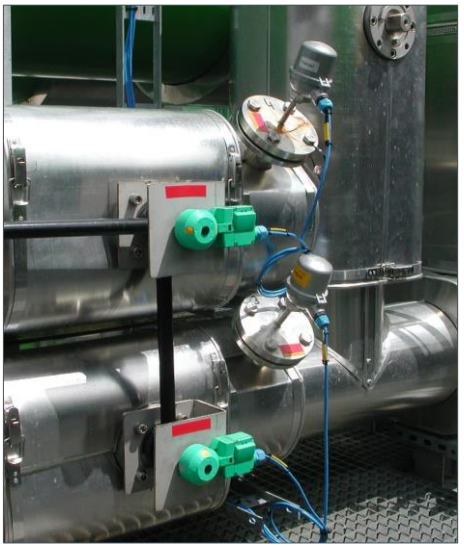
II 1 G

Ex ia IIC T6 Ga

- Category marked without round brackets ( )
- Type of protection marked without squared brackets [ ]
- Temperature class marking
  - → *Installation in the hazardous area*

Typically stated in the documentation:

- Max. input voltage U_i
- Max. input current I_i
- Max. input power P_i
- Effective internal inductance L_i
- Effective internal capacitance C_i



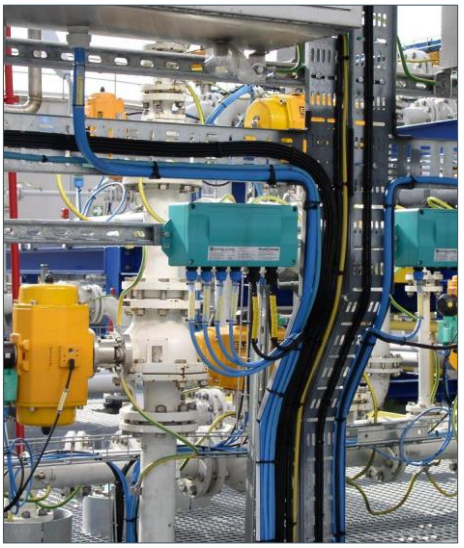
Cables and Energy Storages

Cables are energy storages defined by:

- Inductance per unit length L' (e.g. in mH/m)
- Capacitance per unit length C' (e.g. in nF/m)
  - → *To be determined by manufacturer's information, measurement of a sample or generic values from EN 60079-14*

Prevention of inadmissible increase of energy:

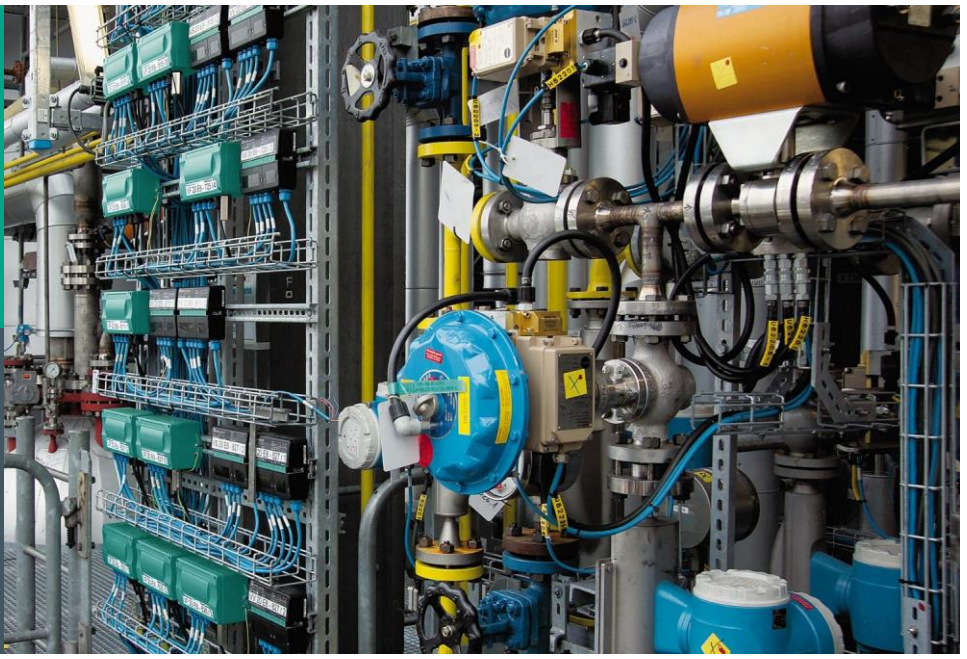
- Protection from electromagnetic coupling
- Protection from damage and corrosion and thus galvanic coupling



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2. Selection of Equipment

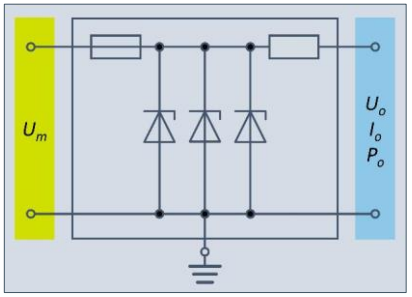
EPL, Zones and Categories



Selection of Equipment acc. to EN 60079-14

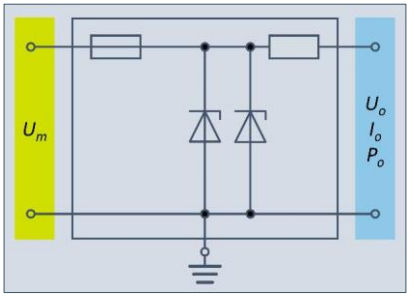
Required EPL	Group	Circuit	„typical“ zone	Category acc. to 2014/34/EU
Ga	II	ia	0	1G
Gb		ia or ib	1	1G or 2G
Gc		ia, ib or ic	2	1G, 2G or 3G
Da	III	ia	20	1D
Db		ia or ib	21	1D or 2D
Dc		ia, ib or ic	22	1D, 2D or 3D

Examples: Zener Barriers ia, ib and ic



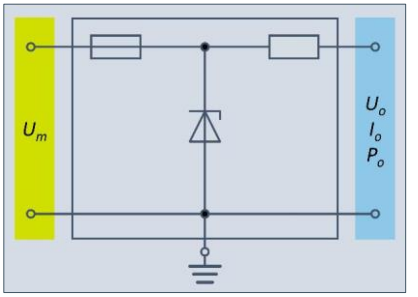
Conventional Zener barrier, ia:

- 2-failure safe regarding intrinsic safety
  - → *Corresponding to category 1 acc. to directive 2014/34/EU*
  - → *Suitable for zone 0; 20*



Conventional Zener barrier, ib:

- 1-failure safe regarding intrinsic safety
  - → *Corresponding to category 2 acc. to directive 2014/34/EU*
  - → *Suitable for zone 1; 21*



Conventional Zener barrier, ic:

- 0-failure safe regarding intrinsic safety
  - → *Corresponding to category 3 acc. to directive 2014/34/EU*
  - → *Suitable for zone 2; 22*

Examples

Combination of:

- Isolated switch amplifier: Ex II (1) G [Ex ia Ga] IIC
- Inductive proximity sensor: Ex II 1 G Ex ia IIC T6 Ga
  - → *Resulting category/EPL?*
  - → *Resulting group?*

Combination of:

- Transmitter supply: Ex II (1) G [Ex ia Ga] IIC
- Temperature sensor: Ex II 1 G Ex ia IIB T4 Ga
  - → *Resulting category/EPL?*
  - → *Resulting group?*

Examples

Combination of:

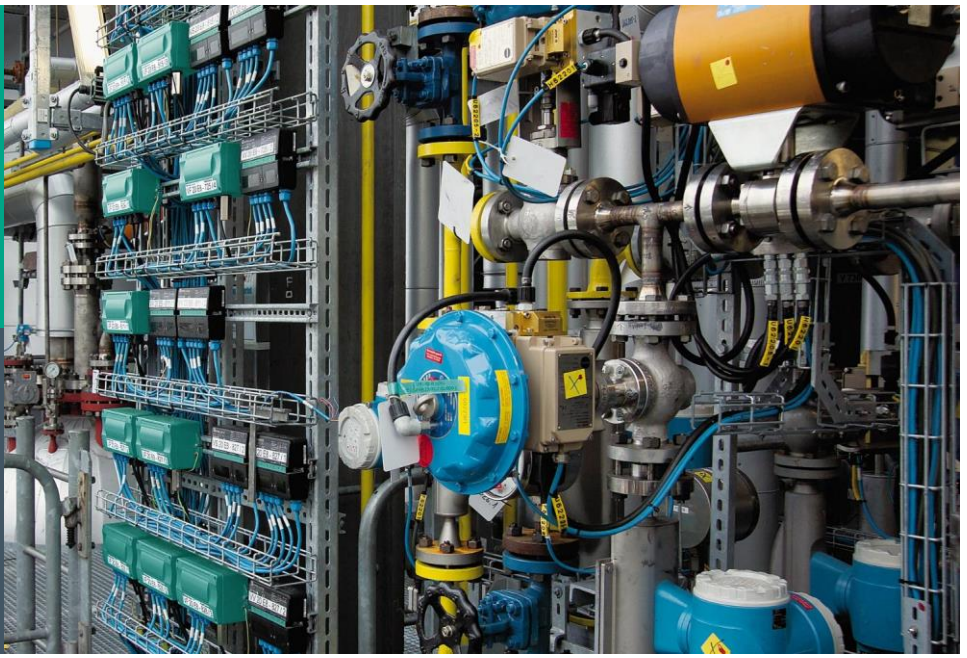
- Solenoid driver: II (1) G [Ex ia Ga] IIA
- Solenoid: II 2 G Ex ia IIC T6 Gb
  - → *Resulting category/EPL?*
  - → *Resulting group?*

Combination of:

- Zener barrier: II (1) G [Ex ia Ga] IIC
- Smoke detector 1: II 2 G Ex ib IIB T4 Gb
- Smoke detector 2: II 2 G Ex ib IIB T3 Gb
  - → *Resulting category/EPL?*
  - → *Resulting group?*
  - → *Resulting temperature class?*

3. Verification of Intrinsic Safety

Procedure, Effects on Maintenance Work



Verification of Intrinsic Safety



istockphoto

The main aspects regarding the verification are:

- Are ignition capable sparks avoided?
- Are inadmissibly hot surfaces avoided?
  - → *The verification is usually carried out by calculation*
  - → *Acc. to (international, European, national) standards the system designer is responsible for the verification*

Procedure: Comparison of U, I, P, L, C

Associated apparatus		cable		Intrinsically safe apparatus
U_o	≤			U_i
I_o	≤			I_i
P_o	≤			P_i
L_o	≥	L_c	+	L_i
C_o	≥	C_c	+	C_i

Verification of intrinsic safety:

- The associated apparatus' output values U_o, I_o and P_o must not exceed the intrinsically safe apparatus' maximal input values
- Cable and safe apparatus reactances must not exceed the maximum values L_o and C_o
  - → *Usually different L_o and C_o values are documented for IIA, IIB and IIC*

Example: Simple Circuit



Associated apparatus			Cable			Intrinsically safe apparatus		
U_o =	26,2	V	ℓ =	200	m	U_i =	28	V
I_o =	93	mA	L' =	1	mH/km	I_i =	100	mA
P_o =	634	mW	C' =	200	nF/km	P_i =	800	mW
L_o =	4,11	mH	L_c =	0,2	mH	L_i =	2	mH
C_o =	92	nF	C_c =	40	nF	C_i =	50	nF



Intrinsically Safe Circuit Maintenance



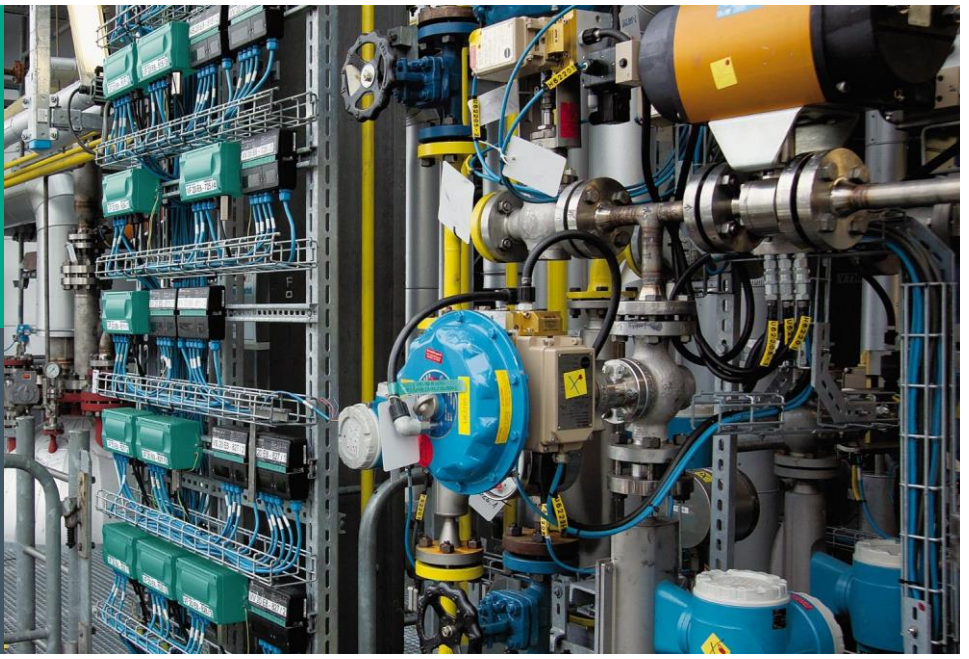
istockphoto

Take care when replacing associated or intrinsically safe equipment:

- The effects on the verification of intrinsic safety must be checked
- The circuit may function technically, but the verification may be impaired!

4. Installation Requirements

Cables, Terminals, Conductors, Shielding, Earthing



Installation Philosophy

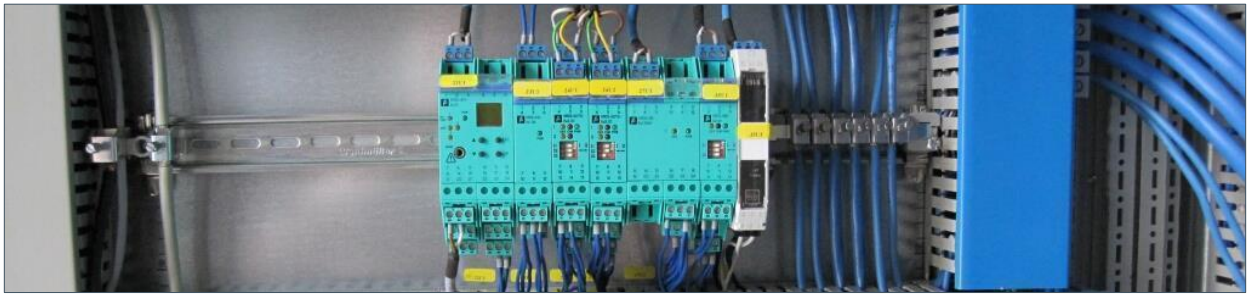


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Particularity of intrinsically safe circuits :

- In case of other types of protection sparks are usually not admissible
- In intrinsically safe circuits sparks are admissible even in normal mode of operation
  - → *The integrity of intrinsically safe circuits has to be protected from the intrusion of energy by separation from other circuits*

Installation of Associated Apparatus



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Associated apparatus are usually installed in the non-hazardous area:

- Non-intrinsically safe parts of associated apparatus (e.g. contacts of relay outputs, hot components etc.) have to be considered as potential sources of ignition
  - → *Exception: the associated apparatus is protected by an additional, appropriate type of protection*

Associated Apparatus in “d” Enclosures

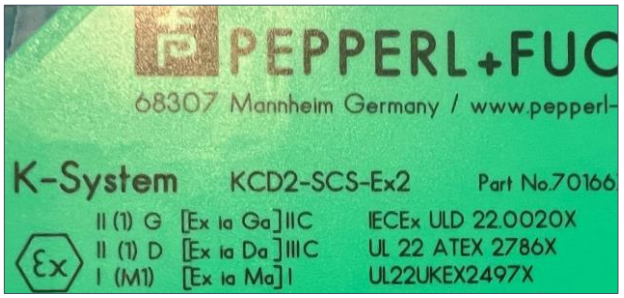


istockphoto

Attention when mounting an associated apparatus in flameproof enclosures “d”:

- Alteration of the internal components of a flameproof enclosure certified as equipment requires a re-evaluation (pressure piling effect!)
- Enclosures with a component certificate only („U“) shall not be installed without certification of the complete assembly (with or without “X”)
  - → *Manufacturer’s responsibilities acc. to 2014/34/EC*

Barriers in Zone 2



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II (1) G [Ex ia Ga] IIC

- Installation in the non-hazardous area only

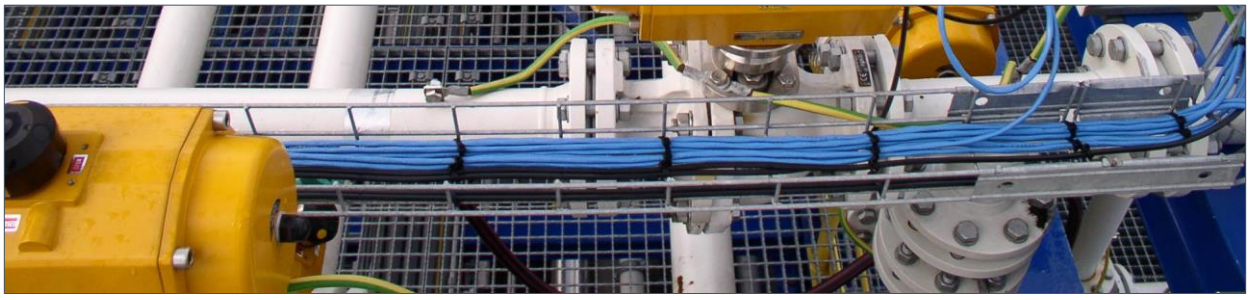


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II 3 G Ex ec IIC T4 Gc ... X

- Installation in the non-hazardous area or
- Installation in zone 2
  - → *Attention should be paid to the "X"!*

Cables for Intrinsically Safe Circuits

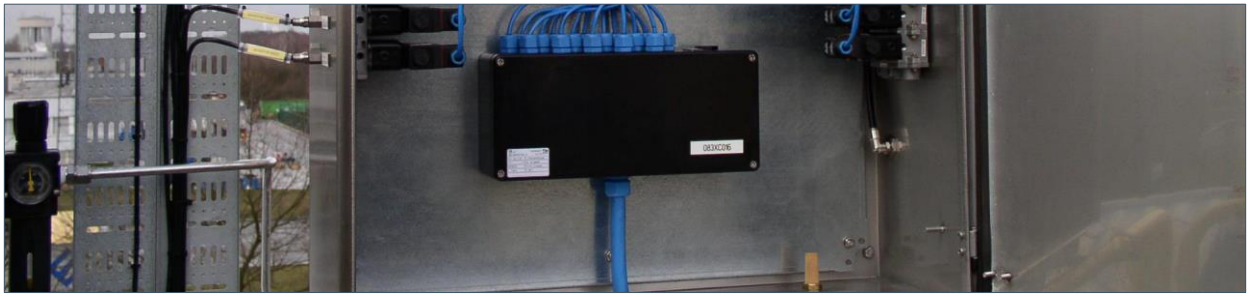


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The following minimum requirements shall be fulfilled:

- Rated insulation voltage (conductor-earth, conductor-screen, screen-earth):
  ≥ AC 500 V or ≥ DC 750 V respectively

Additional Requirements: Multi-Core Cables



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In case of more than one circuit in a cable:

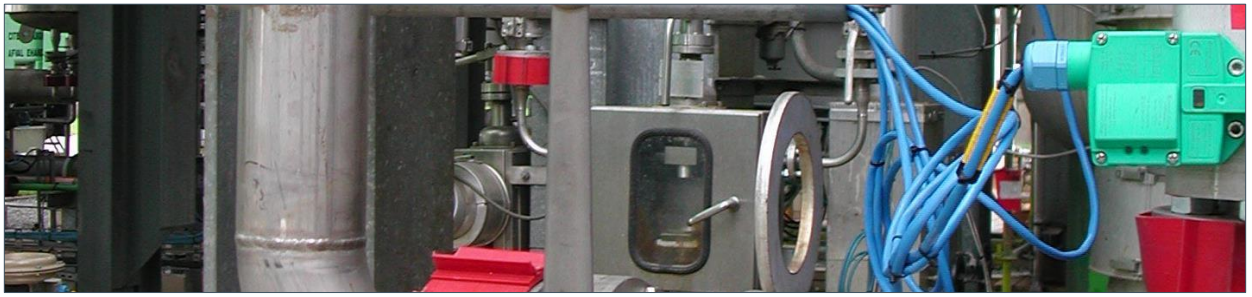
- Intrinsically safe and non-intrinsically safe circuits shall not be carried in one cable
- The radial thickness of the conductor insulation shall be at least 0,2 mm
  - → *Differentiation between type A, type B and type C cables*

Type A, Type B and Type C Cables

Cable	Requirement	Insulation test	Remark
Type A	▪ Individual screen for each intrinsically safe circuit required ▪ Screen coverage of at least 60 % required	▪ AC 500 V or DC 750 V between all joined screens/armours and all joined conductors	No fault consideration required
Type B	▪ Fixed cable installation ▪ Protection from damage ▪ No associated apparatus with U_o > 60 V	▪ AC 500 V or DC 750 V between all joined screens/armours and all joined conductors ▪ AC 1000 V or DC 1500 V between the one half of the joined conductors and the other half	No fault consideration required
Type C	▪ Cable is neither type A nor type B		Consideration of 2 short-circuits + 4 open-circuits simultaneously (ia, ib)

Intrinsically safe “ic” circuits are permitted to be run together with intrinsically safe “ia” and “ib” circuits provided they are run in a cable of type A or type B

Protection of Cables



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Protection from electromagnetic coupling can be achieved by:

- Twisted-core cables
- Screened cables
- Distance between intrinsically safe circuits and sources of interference
  - → *Identification of the appropriate measure by a risk analysis*

Marking of Cables



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Cables of intrinsically safe circuits shall be identifiable:

- Appropriate marking is required
- If marking is carried out by colour it shall be light blue
  - → *In this case blue colour shall not be used for other purposes in a manner which could lead to confusion*
  - → *Recommendation: continuous marking of the cable*

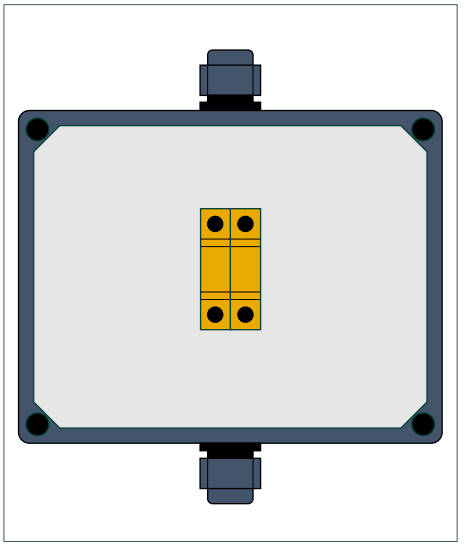
Junction Box – Non-Intrinsically Safe Circuit

Installation in the non-hazardous area:

- Certification not required
- Min. IP2X

Installation in the hazardous area:

- Certification required (typ. „e“, „d“, „t“)
- Min. IP2X



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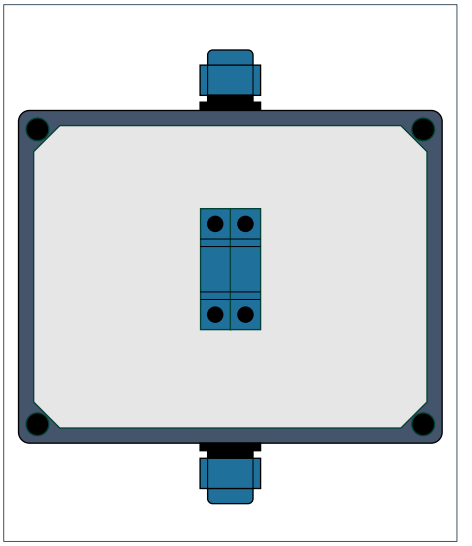
Junction Box – Intriniscally Safe Circuit

Installation in the non-hazardous area:

- Certification not required
- Min. IP2X
- 3 mm distance to earthed parts

Installation in the hazardous area:

- Min. IP2X
- 3 mm distance to earthed parts
- Plastic enclosures: static electricity
- Metal enclosures: static electricity and light metal concentrations



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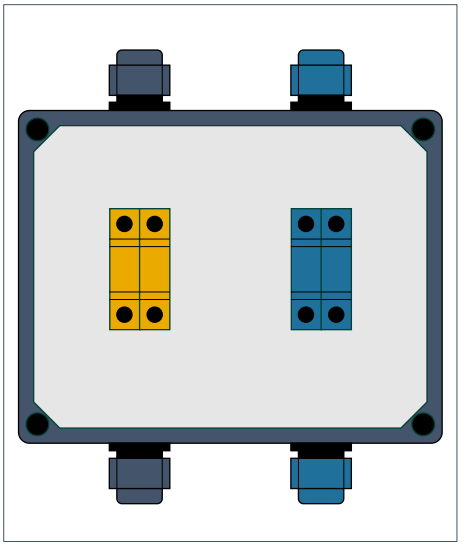
Junction Box – Intriniscally And Non-Intrinsically Safe Circuit

Installation in the non-hazardous area:

- Certification not required
- Min. IP2X
- 3 mm distance to earthed parts (“i”)
- 6 mm distance between circuits „i“
- 50 mm distance between „i“ and non „i“

Installation in the hazardous area:

- Certification required (typ. „e“, „d“, „t“)
- 3 mm distance to earthed parts (“i”)
- 6 mm distance between circuits „i“
- 50 mm distance between „i“ and non „i“



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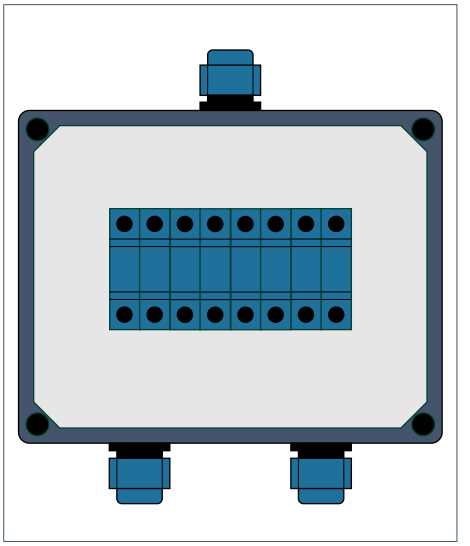
Junction Box – Multiple Intrinsically Safe Circuits

Installation in the non-hazardous area:

- Certification not required
- Min. IP2X
- 3 mm distance to earthed parts (“i”)
- 6 mm distance between circuits „i“

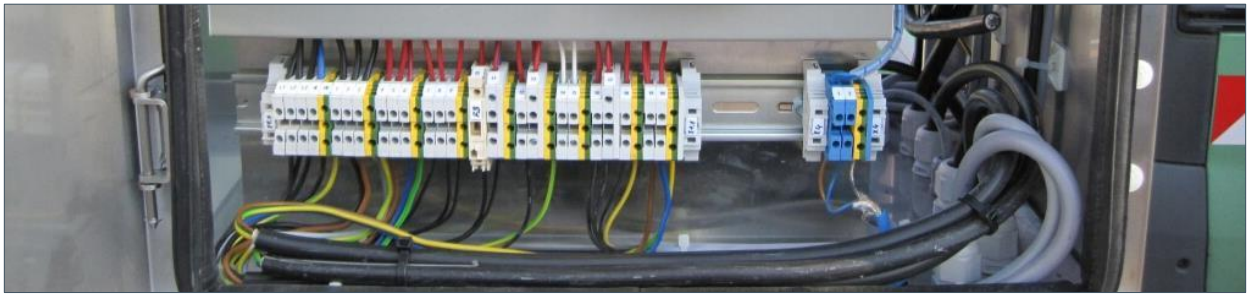
Installation in the hazardous area:

- Certification required
- Min. IP2X
- 3 mm distance to earthed parts (“i”)
- 6 mm distance between circuits „i“
- Plastic enclosures: static electricity
- Metal enclosures: static electricity and light metal concentrations
- Dust, group IIIC: min. IP5X



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Intrinsically Safe Terminal Requirements

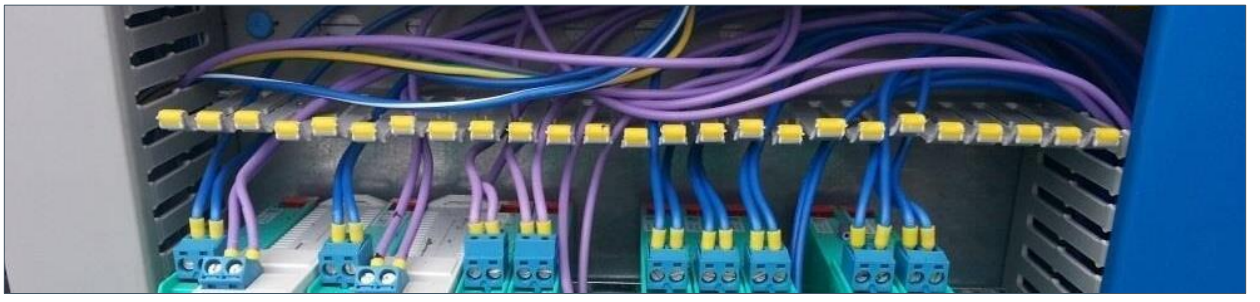


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Terminals of intrinsically safe circuits shall be segregated from:

- Terminals of non-intrinsically safe circuits (50 mm)
- Terminals of other intrinsically safe circuits (6 mm)
- Earthed or other conductive parts (3 mm)

Installation of Conductors



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Installation of intrinsically safe conductors in cabinets, boxes etc.:

- Separation from other conductors by an intermediate layer (insulating or earthed metal)
  - → *Exception: intrinsically safe or other circuits are screened or metal sheathed*

Earthing of Intrinsically Safe Circuits



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Connection of intrinsically safe circuits to earth:

- Intrinsically safe circuits shall either be isolated from earth or connected to the equipotential system at one point only
  - → *Earthing is required if barriers without galvanic isolation (Zener barriers) are installed*